

API Contract Document

IntelliPlant AI

API Specifications

Version: 1.0

Base URL

```
/api/v1
```

Authentication APIs

POST /auth/login

Request

```
{  
  "email": "user@example.com",  
  "password": "password"  
}
```

Response

```
{  
  "access_token": "",  
  "refresh_token": "",  
  "role": "Engineer"  
}
```

POST /auth/register

Request

```
{
  "name": "Narayan",
  "email": "abc@gmail.com",
  "password": "*****"
}
```

Document APIs

POST /documents/upload

Multipart Upload

Response

```
{
  "document_id": "",
  "status": "uploaded"
}
```

GET /documents

Response

```
[
  {
    "id": "",
    "filename": "manual.pdf"
  }
]
```

DELETE /documents/{id}

Deletes document.

Chat APIs

POST /chat

Request

```
{
  "query": "Why Pump P101 failed?"
}
```

Response

```
{
  "answer": "",
  "citations": [
    "manual.pdf"
  ],
  "confidence": 0.93
}
```

Search APIs

POST /search

Request

```
{
  "query": "Bearing failure"
}
```

Response

```
{
  "results": []
}
```

RCA APIs

POST /rca

Request

```
{
  "equipment": "Pump P101"
}
```

Response

```
{
  "root_cause": "",
  "recommendation": ""
}
```

Compliance APIs

POST /compliance/check

Response

```
{
  "violations": [],
  "recommendations": []
}
```

Lessons Learned APIs

POST /lessons

Response

```
{
  "patterns": [],
}
```

```
"actions": []
}
```

Knowledge Graph APIs

GET /graph/{equipment_id}

Response

```
{
  "nodes": [],
  "edges": []
}
```

Analytics APIs

GET /dashboard/metrics

Response

```
{
  "documents": 100,
  "incidents": 20,
  "equipment": 50
}
```

WebSocket

```
/ws/chat
```

Real-time agent streaming.

Conclusion

REST APIs expose all platform capabilities while supporting future GraphQL and gRPC expansion.